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Real-Time Multimodal Detection of Fraud and Scam Calls Using Conversation Stage Modeling and Ensemble Learning

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Subhalaxmi Chakraborty[1], Krishna Kumar Jha[2], Pritam Paul[1], Arya Gupta[3], Srijan Sarkar[3], Shinjan Saha[3]

[1]Department of CST & CSIT, Institute of Engineering & Management, University of Engineering & Management, Kolkata, India

[2]Guru Nanak Institute of Technology, Kolkata, India

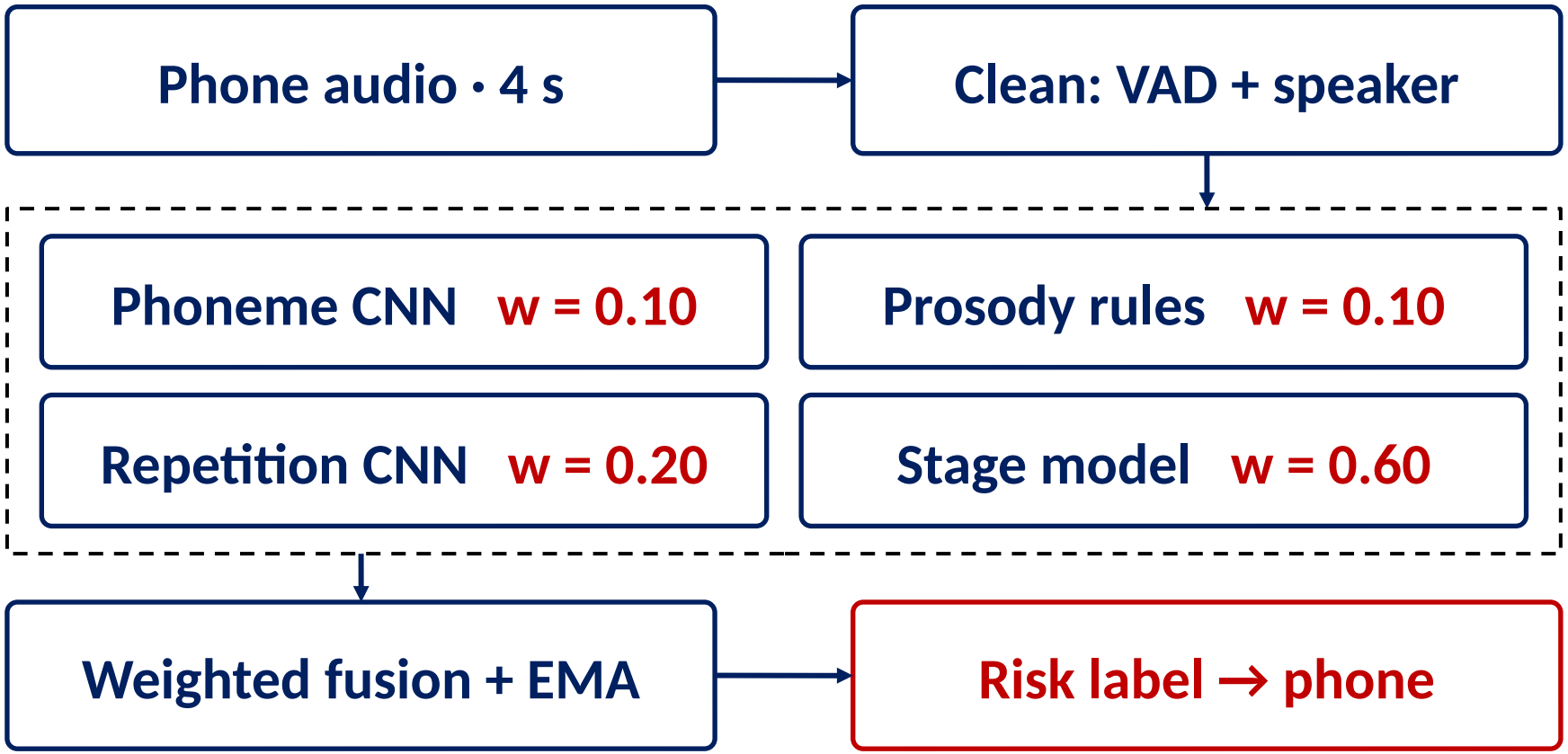
[3]Department of CSE, Institute of Engineering & Management, University of Engineering & Management, Kolkata, India

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Objectives

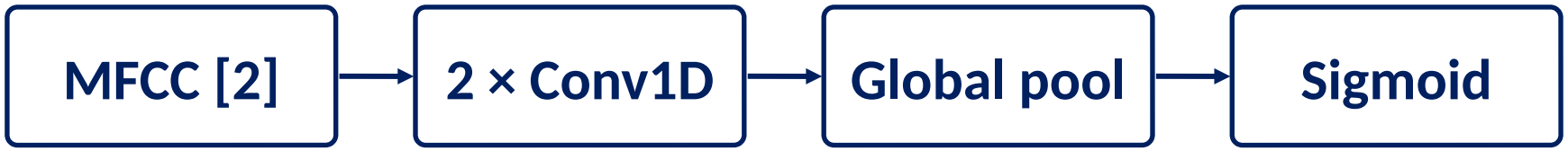
- Scammers persuade, not hack: harm happens **during the call**
- Scam calls follow a script [1]:
  - Greeting → Authority → Problem → Urgency → Data
- **Goals**
  - Warn the user live, every 4 seconds
  - Track the stage of the conversation
  - Fuse voice cues and meaning in one ensemble
  - Keep false alarms low
  - Deliver a working Android app and server

Proposed Framework



All four models run in parallel on every window

Acoustic Models (CNNs)



- **Phoneme CNN [3]**
  - Input 370 × 120
  - ≈ 80.6 k parameters
  - Phoneme sounds
- **Repetition CNN [4]**
  - 4 × 1-s segments
  - ≈ 17.2 k parameters
  - Scripted phrases

$$h_t = \mathrm{ReLU}\left(b + \sum_{k=0}^{K-1} W_k \, x_{t+k} \right)$$

$$p_{\mathrm{rep}} = \max_{j=1,\ldots,4} \mathrm{CNN}(s_j)$$

Prosody and Stage Models

Prosody: urgency score

$$u = \min\left(3,\ \sum_{i=1}^{5} \mathbf{1}[f_i > \lambda_i\, b_i]\right)$$

5 cues: pitch, pitch range, loudness, brightness, speed

Conversation stage [5], [6]

Whisper → MiniLM → cosine ≥ 0.55 → forward only

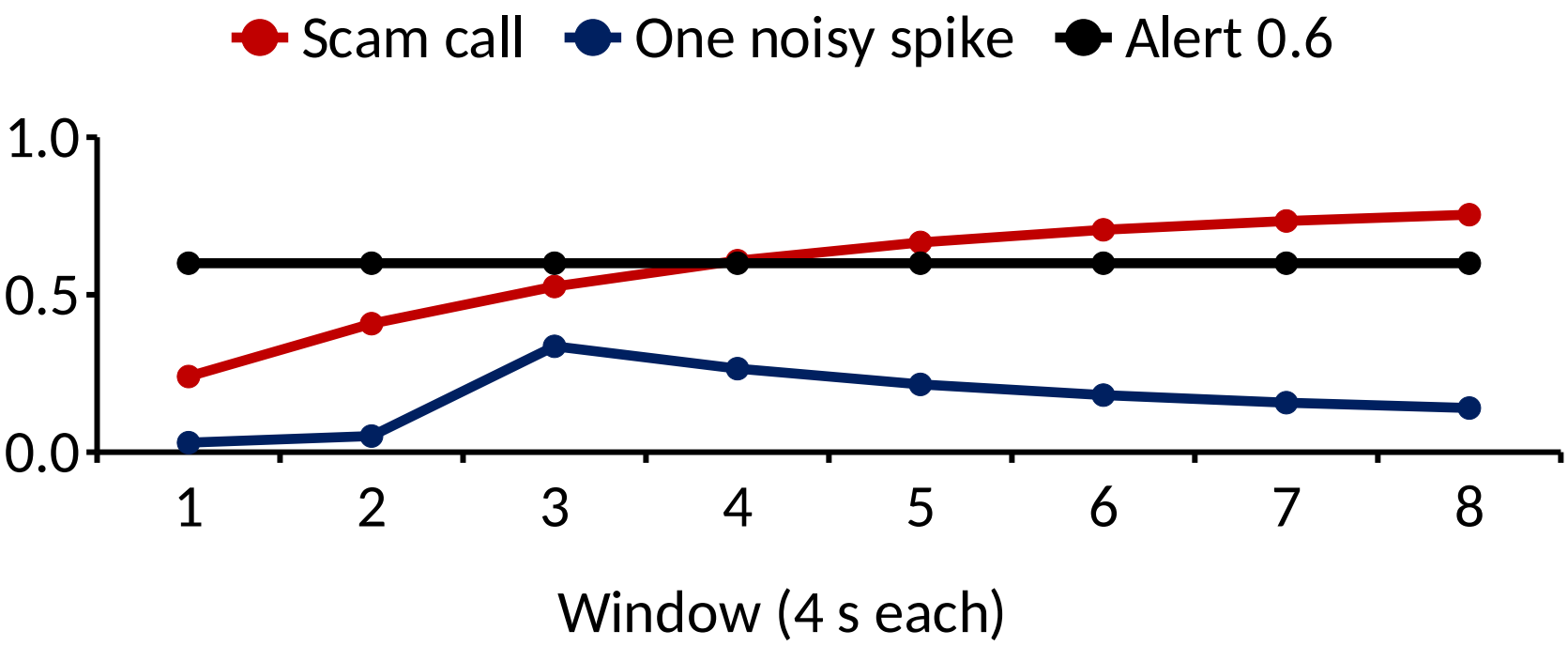


Risk rises with each stage and never goes back

Fusion and Smoothing

$$r_t = 0.6\, r_{\text{stage}} + 0.2\, p_{\text{rep}} + 0.1\, p_{\text{phon}} + 0.1\, r_{\text{urg}} \quad [7]$$

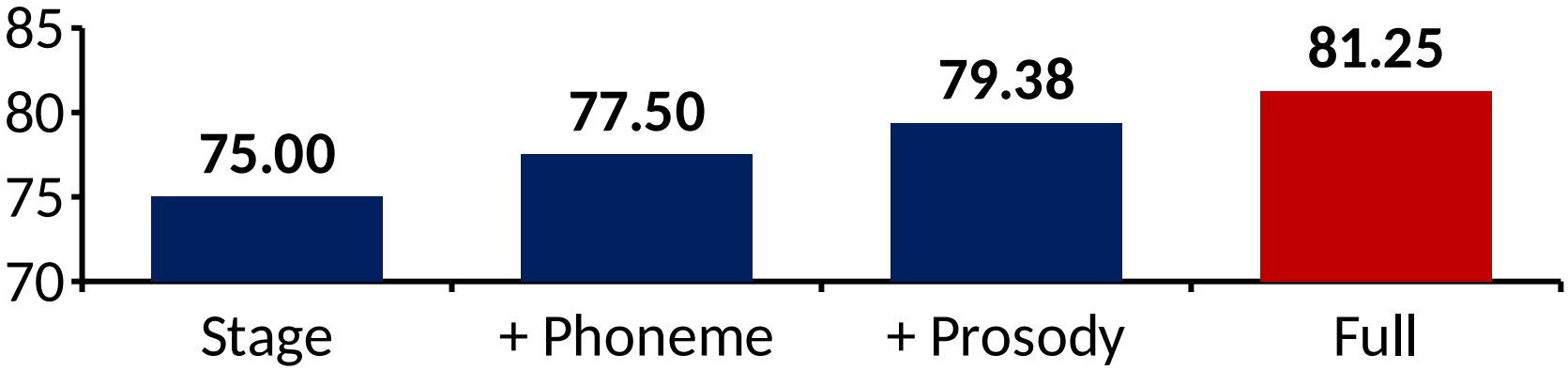
$$R_t = 0.7\, R_{t-1} + 0.3\, r_t$$



Results and Conclusion

81.25%	0.81	0.873	0.875
Accuracy	F1-score	ROC-AUC [8]	Precision

Ablation on 16 test calls: accuracy (%)



- Stage tracking + 3 voice cues → stable live warning
- Next: larger multilingual data, on-device models

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Thank You · Questions?